

Calibration Report

Human - IH_powerLaw_stressBased

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Excellent

QUALITY



1.00

WORST R-HAT



5000

MIN ESS



Run Configuration

Species	human
Model	IH_powerLaw_stressBased
Control variables	shear_stress, exposure_time
Likelihood	normal
MCMC walkers	100
Burn-in steps	2500
Production steps	2500
Calibrate noise	Yes

PRIORS

PARAMETER	DISTRIBUTION	BOUND / LOCATION	BOUND / SCALE
A	Uniform	1e-07	20.0
alpha	Uniform	0.5	3.0
beta	Uniform	0.01	1.0
sigma	Uniform	1e-09	0.5

POSTERIOR SUMMARY

PARAMETER	MEDIAN (68% CI)	MODALITY
A	10.8733 (+0.9995 / -0.8675)	Unimodal
alpha	0.8942 (+0.1815 / -0.1587)	Unimodal
beta	0.2140 (+0.0753 / -0.0686)	Unimodal
sigma	0.0908 (+0.0080 / -0.0070)	Unimodal

MCMC Diagnostics

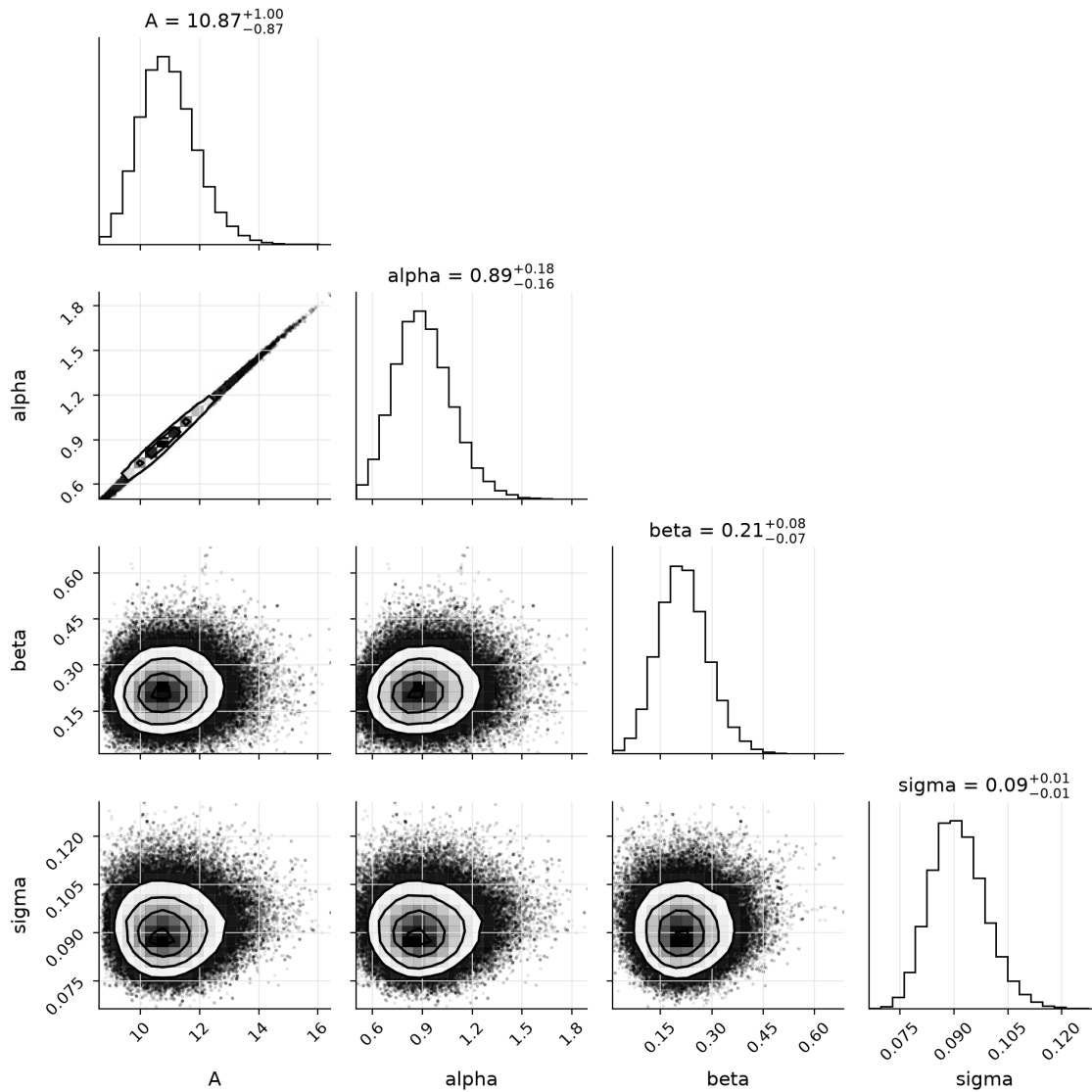


Figure 1 - Pairwise posterior corner plot (post burn-in).

Table 1 - Convergence diagnostics (post burn-in, 100 walkers).

PARAMETER	ESS BULK	ESS TAIL	R-HAT	ESS QUALITY	R-HAT QUALITY
A	5000	9300	1.00	Excellent	Excellent
alpha	5000	9400	1.00	Excellent	Excellent
beta	5100	9700	1.00	Excellent	Excellent
sigma	5400	11000	1.00	Excellent	Excellent

R-hat thresholds: < 1.01 = Excellent, 1.01-1.10 = Good, >= 1.10 = Poor (Vehtari et al., 2021).

ESS thresholds: > 400 = Excellent, 101-400 = Acceptable, <= 100 = Poor (Vehtari et al., 2021).

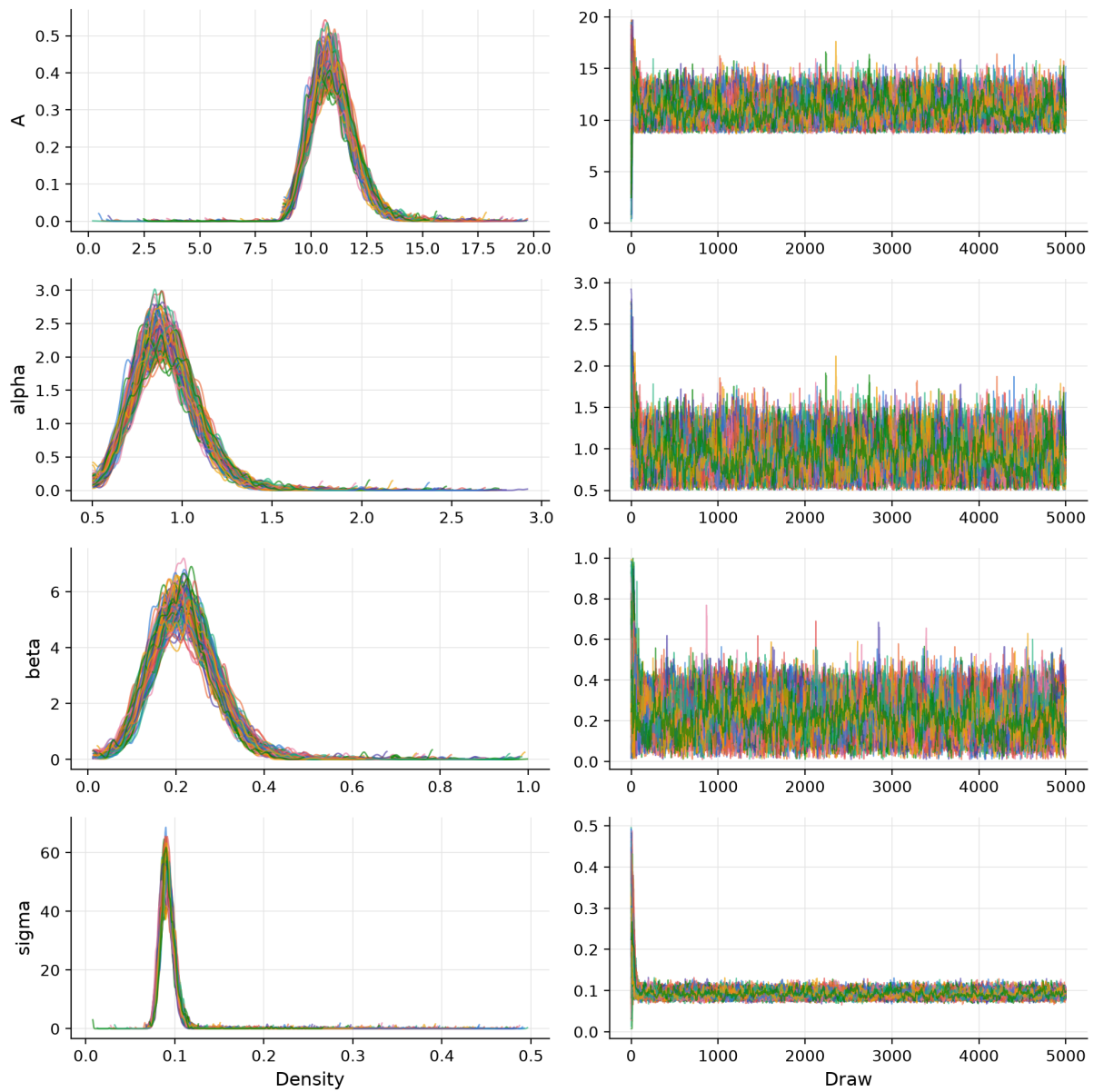


Figure 2 - Trace plots. Left: posterior density (full chain); right: sample traces per walker.

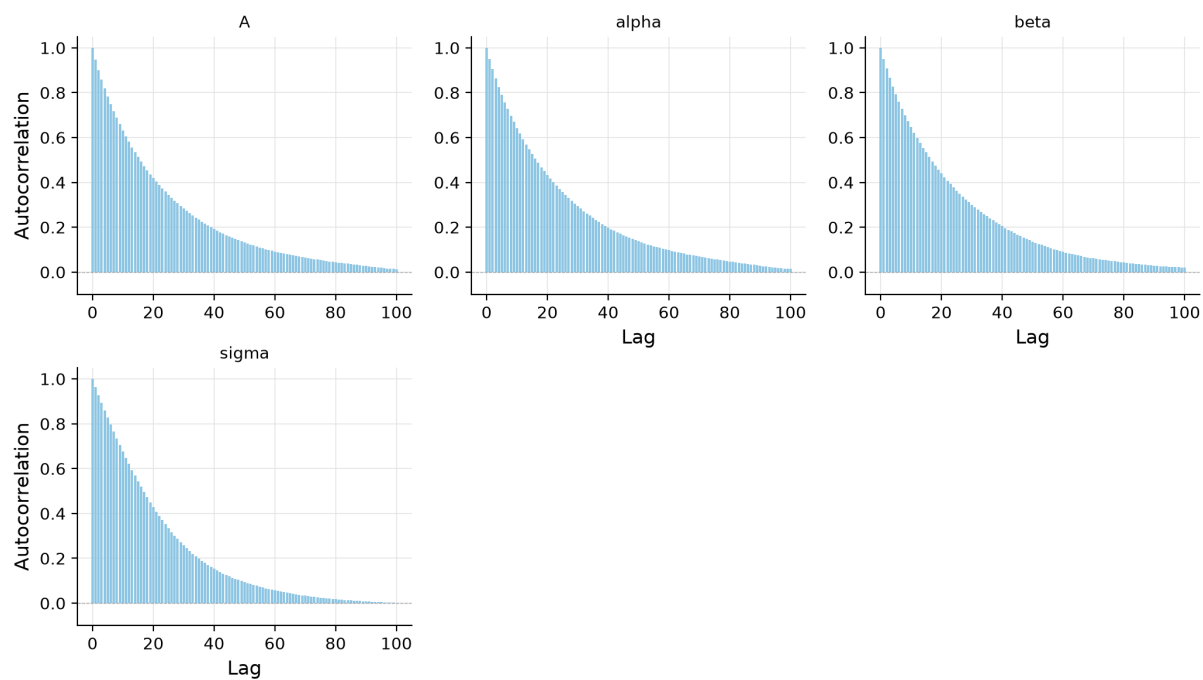


Figure 3 - Autocorrelation by lag for each parameter.